

Careless Responding Detection: The Field, Current Practices, and Software

Vittorio Guerrieri
Chapter 1 of the thesis

The Problem of Careless Responding

Self-report questionnaires rest on a tacit assumption: that respondents read each item, retrieve the relevant information, form a judgment, and map it onto the response scale. When this process breaks down—when respondents answer without attending to item content—the resulting data carry the surface form of valid measurement but none of its substance. This phenomenon, variously labeled *careless responding*, *insufficient effort responding* (IER), *inattentive responding*, or *random responding*, has become one of the most actively studied threats to the validity of survey research (Curran, 2016; Meade & Craig, 2012; Ward & Meade, 2023).

The terminological proliferation reflects subtly different emphases. Huang et al. (2012) introduced *insufficient effort responding* to foreground the motivational mechanism: respondents who are unwilling to expend the cognitive effort the task demands. Krosnick (1991) had earlier framed the same behavior as *satisficing*—respondents who, rather than *optimizing* their answer through the full cognitive sequence, settle for a response that is merely good enough, or in the limiting case of *strong* satisficing, select responses with little or no reference to item content. *Careless responding* is the broader umbrella term now favored in methodological reviews (Ward & Meade, 2023); it encompasses both the low-effort respondent and the respondent whose attention lapses intermittently over a long instrument. Throughout this thesis I use *careless responding* as the cover term and reserve more specific labels for particular response patterns.

Whatever the label, the empirical consequences are well documented and consistently damaging. Careless responding inflates measurement error and depresses scale reliability (Huang et al., 2012). Because careless responses are, to a first approximation, unrelated to the constructs a questionnaire targets, they attenuate true relationships between variables; yet they can also *create* spurious relationships, most insidiously when careless respondents endorse the midpoint or an extreme of every scale and thereby manufacture artifactual covariance across otherwise unrelated measures (Huang et al., 2015). Maniaci and Rogge (2014) demonstrated across several studies that even modest proportions of inattentive participants can meaningfully distort effect sizes, suppress genuine effects, and reduce statistical power. Kam and Meyer (2015) showed that careless and acquiescent responding can distort the very factor structure recovered from a set of items, splitting a unidimensional construct into spurious method factors. In clinical and applied settings the stakes are higher still: Conijn et al. (2019) found that careless responding in routine outcome monitoring biased symptom estimates in ways that could affect treatment decisions.

Estimates of prevalence vary with population, mode of administration, and detection method, but they are rarely negligible. Reviews place the typical incidence somewhere between roughly 5% and 15% of respondents in well-controlled samples, with substantially higher rates in unsupervised online panels and crowdsourced labor markets (Curran, 2016; Meade & Craig, 2012; Ward & Meade, 2023). Bowling et al. (2016) added an important nuance: carelessness is not purely situational noise but is partly a stable individual difference, correlated with low conscientiousness and low agreeableness, meaning that the same respondents tend to provide low-effort data across studies. The practical upshot, emphasized in every recent treatment of the topic, is that careless responding cannot be assumed away: it must be actively prevented where possible and detected and addressed where not (DeSimone & Harms, 2018; DeSimone et al., 2015; Ward & Meade, 2023).

A Taxonomy of Causes and Response Patterns

Effective detection requires a clear picture of *what careless responding looks like* in the data, because different careless behaviors leave different statistical fingerprints. The literature converges on a useful organizing distinction between *consistent* and *inconsistent* careless responding (Curran, 2016; Ward & Meade, 2023).

Consistent careless patterns are those in which the respondent produces a regular, low-information sequence. The clearest example is *straightlining* (also called *non-differentiation*): selecting the same response option for a long run of consecutive items, often the scale midpoint or an extreme. A closely related notion is the *longstring*—the maximum length of consecutive identical responses, which Johnson (2005) established as a simple but effective marker of inattentive protocols. *Acquiescent* responding (uniform agreement regardless of item keying) is another consistent pattern, particularly damaging because it survives naive reliability checks while corrupting the covariance structure (Kam & Meyer, 2015). Consistent patterns are, paradoxically, both easy and hard to detect: easy because their regularity is conspicuous to a longstring counter, hard because a respondent who endorses the midpoint of every item can look indistinguishable from a genuinely neutral one.

Inconsistent careless patterns, by contrast, involve responses that vary across items but bear no coherent relationship to item content. The prototype is *purely random* responding, in which answers are effectively drawn at random from the response scale. Such protocols display normal-looking variance and no long runs, so they slip past longstring and variability checks; what betrays them is the *absence of the correlational structure* that attentive respondents exhibit—answers to items measuring the same latent trait, which should covary, instead scatter (Curran, 2016; Meade & Craig, 2012). *Mixed* patterns, in which a respondent attends to part of the instrument and disengages for the rest (for example through mid-survey *fatigue*), combine features of both families and are correspondingly difficult to catch with any single index (Ward & Meade, 2023).

This taxonomy matters for a reason that runs through the entire thesis: the indices in widest use were largely designed for, and excel at, consistent careless patterns, whereas inconsistent patterns—random and mixed responding—are precisely where most methods are weakest. The cross-factor coherence approach developed in later chapters is motivated by this gap.

Prevention by Design

Because no detection method is perfect, the first line of defense is to design instruments and data-collection procedures that discourage careless responding and that embed signals of inattention directly into the questionnaire (DeSimone et al., 2015; Ward & Meade, 2023).

The most common device is the *attention check*, which takes several forms. *Instructed-response items* (IRIs) direct the respondent to select a specific option (“Please select *strongly agree* for this item”); a deviation flags inattention. *Bogus items* assert something patently true or false (“I was born on February 30th”); an implausible answer is diagnostic. *Instructional manipulation checks* (IMCs), introduced by Oppenheimer et al. (2009), embed a comprehension test within a seemingly ordinary item and have been shown to identify respondents who are satisficing rather than reading. A self-report single-item approach simply asks respondents, at the end, whether their data should be used—a surprisingly informative question when posed without penalty (Meade & Craig, 2012).

These instruments are valuable but limited, and the limitations are instructive. Curran and Hauser (2019) showed that attention-check items generate a non-trivial rate of *valid-but-incorrect* responses: attentive participants sometimes fail them (through misreading or genuine ambiguity), while determined careless respondents can pass an isolated check by chance or by briefly attending to a conspicuous item. A single check therefore has limited sensitivity and imperfect specificity. Attention checks also consume questionnaire space, can alter respondent behavior by signaling surveillance, and—being placed at fixed positions—catch inattention only at those points, missing the intermittent disengagement characteristic of fatigue. For time-stamped or computer-administered surveys, *response latency* offers a complementary design-based signal: implausibly fast page completions indicate that a respondent could not have read the items, a logic formalized by Wise and Kong (2005) in the *response time effort* measure and later embedded in model-based approaches (Ulitzsch et al., 2022). Response times, however, are unavailable in the many settings where only the final responses are recorded.

The recurring lesson is that prevention reduces but does not eliminate careless responding, and that the design-based signals it provides are partial. This motivates *post-hoc* detection from the response data themselves—methods that require no special items, no timing information, and no advance planning, and that can therefore be applied to data already collected.

Post-hoc Identification Methods

Post-hoc methods infer carelessness from the structure of the observed responses. They fall into several families, each exploiting a different statistical signature of inattention. The two most authoritative syntheses of this literature—Meade and Craig (2012) and Curran (2016)—organize the field around a core set of indices that I summarize here and collect, with their software implementations, in Table 1.

Consistency and Response-Pattern Indices

Longstring counts the maximum number of consecutive identical responses in a protocol. It is the canonical detector of straightlining and is trivial to compute, but it is

blind to inconsistent careless responding and is biased by the number of items keyed in the same direction (Johnson, 2005; Meade & Craig, 2012).

The *intra-individual response variability* (IRV) is the standard deviation of a respondent’s answers across a set of items (Dunn et al., 2018). Very low IRV signals straightlining (little or no variation), while implausibly high IRV can indicate random responding across the full scale range. Because it condenses a protocol to a single dispersion statistic, IRV is cheap and intuitive, but it discards all information about *which* items were answered how.

Psychometric synonyms and antonyms (Meade & Craig, 2012) operationalize coherence at the item-pair level. One identifies pairs of items that are strongly positively correlated in the sample (synonyms) or strongly negatively correlated (antonyms), then computes, *within each respondent*, the correlation between their answers to the two members of each pair. An attentive respondent answers synonym pairs similarly and antonym pairs oppositely, producing a high within-person correlation; a careless respondent does not. This is a powerful idea because it targets inconsistent careless responding directly, and it is the conceptual ancestor of the cross-factor coherence method developed later in this thesis.

The *person-total correlation* correlates a respondent’s profile of answers with the sample’s mean response to each item; respondents who answer against the grain of the typical pattern receive low values. *Even-odd consistency* splits a scale into even- and odd-numbered items, computes two subscale scores, and assesses their agreement; *resampled individual reliability* generalizes this by repeatedly splitting and correlating. These consistency indices trace back to classical work on individual-protocol validity (Jackson, 1976; Johnson, 2005).

Multivariate Outlier Detection

A complementary strategy treats careless protocols as multivariate *outliers*. The *Mahalanobis distance* (Mahalanobis, 1936) measures how far a respondent’s response vector lies from the sample centroid, scaled by the inverse covariance matrix, and is the most widely used outlier index in this literature (Curran, 2016; Meade & Craig, 2012). Under multivariate normality the squared distance follows a chi-square distribution with degrees of freedom equal to the number of items, which suggests a principled cutoff. In practice, Likert data violate the normality assumption badly, so the chi-square reference distribution is only approximate, and—as the simulation chapters of this thesis document in detail—the nominal cutoff can be far too conservative for questionnaire data, flagging almost no one. Mahalanobis distance also detects *any* multivariate outlier, careless or not, including genuinely unusual but attentive respondents.

Person-Fit and Item-Response-Theory Approaches

A distinct tradition grew out of item response theory (IRT), where *person-fit* statistics quantify how well an individual’s response pattern conforms to the fitted measurement model. The l_z statistic and the Rasch *infit/outfit* indices are the best known; Karabatsos (2003) compared thirty-six such statistics for detecting aberrant response patterns, and Meijer et al. (2016) provided a practical guide to applying them in clinical research. Person-fit methods rest on a strong measurement model, which is their strength (a principled probabilistic account of “aberrance”) and their weakness (sensitivity to model misspecification,

and a need for the model to be estimable). More recently, *mixture-model* formulations treat the population as a blend of attentive and careless latent classes and estimate class membership directly; Ulitzsch et al. (2022) combined item responses with response times in a latent-response mixture model that simultaneously identifies and adjusts for careless and insufficient-effort responding.

Machine-Learning and Ensemble Approaches

The newest family abandons the idea of a single hand-crafted index in favor of *learning* a careless/attentive boundary from many indices at once. Schroeders et al. (2022) trained a stochastic gradient-boosting classifier on a battery of careless indicators and reported detection performance exceeding that of any constituent index, demonstrating that the indices carry *complementary* information that a flexible classifier can exploit. This ensemble logic is directly relevant to the present thesis, whose later chapters combine a novel coherence index with established detectors inside a Random Forest (Breiman, 2001). The turn toward machine learning has not gone uncontested. Alfons and Welz (2024) offer an open-science critique, warning that supervised approaches risk overfitting to the idiosyncrasies of the simulated or instructed-careless data on which they are trained, that ground-truth labels for careless responding are inherently uncertain, and that reproducible benchmarks and shared data are largely absent from the field. Their related work on identifying the *onset* of careless responding within a protocol (Welz & Alfons, 2023) illustrates an alternative, model-based direction. These cautions frame the methodological commitments of this thesis—in particular the emphasis on validation across independent simulation draws and on deployment-realistic, label-free calibration.

Cutoffs, Thresholds, and Combining Indices

Computing an index is only half the task; the other half is deciding *where to draw the line*. Every method above yields a continuous score, and converting that score into a binary careless/attentive flag requires a threshold. This seemingly technical step is in fact one of the least settled and most consequential issues in the field.

Two broad strategies exist. The first uses a *theoretical* cutoff derived from a reference distribution—most prominently the chi-square cutoff for Mahalanobis distance—which has the appeal of requiring no tuning but, as noted, rests on distributional assumptions that questionnaire data routinely violate. The second uses an *empirical* cutoff: a fixed quantile of the observed scores, a visual “elbow” in the score distribution, or a value optimized against known labels. Empirical cutoffs adapt to the data but risk either circularity (when tuned on the same labels used to evaluate them) or arbitrariness (when chosen by eye). Ward and Meade (2023) note that the absence of consensus cutoffs is a genuine obstacle to cumulative science, because two analysts applying “the same” index to the same data can reach different conclusions about who is careless.

The comparative literature offers two robust findings that shape best practice. First, no single index dominates across conditions: Niessen et al. (2016) compared several methods on web-questionnaire data and concluded that the best choice depends on the type of careless responding present and on questionnaire characteristics, with different indices catching different careless respondents. Second—and following directly from the first—*combining* in-

dices outperforms relying on any one of them (Curran, 2016; Schroeders et al., 2022). The recommended practice in recent reviews is therefore to compute multiple indices, examine their convergence, and flag respondents identified by several of them (DeSimone et al., 2015; Ward & Meade, 2023). The ensemble and threshold-calibration methods developed in this thesis are a formalization of exactly this advice: they replace ad hoc multi-index agreement rules with a trained classifier and a deployment-realistic operating point.

Software for Careless Responding Detection

The methods reviewed above are only as useful as the tools that implement them, and here the R ecosystem has become the de facto standard for careless responding detection. This section surveys the packages in current use; their indices and the constructs they target are summarized in Table 1.

The central package is `careless` (Yentes & Wilhelm, 2023), which provides a unified implementation of the core response-pattern indices: `longstring` (and its average-run variant), `irv`, `mahad` (Mahalanobis distance, with optional robust covariance estimation), `evenodd` (even-odd consistency), and `psychsyn/psychant` (psychometric synonyms and antonyms). Because it gathers the indices of Meade and Craig (2012) and Curran (2016) behind a consistent interface, `careless` is the package most often cited in applied work and the natural baseline against which new methods are compared.

Several other packages contribute pieces of the workflow. The `responsePatterns` package (Riháček & Gottfried, 2023) implements autocorrelation-based screening for repetitive response patterns (Gottfried et al., 2022), targeting the machine-like regularity that some careless protocols exhibit. For the multivariate-outlier branch, the `performance` package (Lüdtke et al., 2021) exposes `check_outliers`, which computes Mahalanobis (and several robust alternatives) within a tidy model-checking framework, while the venerable `psych` package (Revelle, 2024) supplies the factor-analytic and reliability machinery—exploratory factor analysis, parallel analysis, and scale-reliability estimation—on which several careless indices and most downstream psychometric analyses depend. Mixture-model approaches to careless detection draw on general-purpose finite-mixture engines such as `mclust` (Scrucca et al., 2016) and `mixtools` (Benaglia et al., 2009), which fit the latent attentive/careless class structure used by model-based methods. All of these run within base R (R Core Team, 2024), and supervised-ensemble approaches add a classifier layer—gradient boosting in Schroeders et al. (2022), or a Random Forest in the present work—on top of the index computations these packages provide.

What the ecosystem still lacks is as informative as what it offers. Alfons and Welz (2024) observe that, despite the maturity of the individual-index tooling, the field has no shared, labeled benchmark dataset against which competing detectors can be evaluated on equal terms, and that machine-learning detectors in particular are typically validated only on the simulated data their authors generate. There is, similarly, no standard, well-validated routine for *calibrating* an index threshold without access to ground-truth labels—the very problem the deployment-oriented chapters of this thesis address. Outside R, careless detection is comparatively underserved: commercial survey platforms offer little beyond fixed-position attention checks, and although general-purpose machine-learning libraries in other languages could in principle implement any of these methods, no established, careless-specific toolkit exists outside the R packages surveyed here.

Table 1

Principal careless-responding detection methods, the careless behavior each is best suited to detect, and their implementation in current R software.

Method	What it captures	Target pattern	R implementation
Longstring	Longest run of identical consecutive responses	Consistent (straightlining)	<code>careless::longstring</code>
IRV	SD of a respondent's answers across items	Consistent (low) / random (high)	<code>careless::irv</code>
Psychometric synonyms / antonyms	Within-person correlation over high- $ r $ item pairs	Inconsistent (random, mixed)	<code>careless::psychsyn</code> , <code>psychant</code>
Person-total correlation	Agreement of a profile with the sample mean profile	Inconsistent	<code>careless</code> / base R
Even-odd / resampled reliability	Internal consistency across item splits	Inconsistent	<code>careless::evenodd</code>
Mahalanobis D^2	Multivariate distance from the sample centroid	Multivariate outliers	<code>careless::mahad</code> ; <code>performance::check_outliers</code>
Autocorrelation screening	Machine-like repetitiveness of a protocol	Consistent (patterned)	<code>responsePatterns</code>
Person-fit (l_z , in-fit/outfit)	Departure from a fitted IRT model	Aberrant patterns	<code>mirt</code> / <code>PerFit</code>
Mixture / response-time models	Latent attentive vs. careless classes	All (model-based)	<code>mclust</code> , <code>mixtools</code>
Ensemble / machine learning	Learned combination of many indices	All	gradient boosting; Random Forest

Gaps in the Field and the Contribution of This Thesis

This review motivates the work that follows by making three gaps explicit.

First, the methods in widest use are strongest on *consistent* careless responding and weakest on the *inconsistent* patterns—random and mixed—that are arguably the hardest to prevent and the most damaging to covariance-based analyses. Longstring and IRV detect regularity; Mahalanobis distance detects outlyingness; but the cross-item *coherence* that distinguishes an attentive from a randomly responding participant is captured only partially, by the synonym/antonym indices, and only for short lists of hand-selected item pairs. A method that exploits the full correlational structure of a multi-construct questionnaire to detect inconsistent careless responding would fill a real gap. This is the motivation for the permutation-based coherence index, *ReReReRe*, developed in Chapter 3.

Second, the field has converged on the principle that *combining* indices beats any single one (Curran, 2016; Niessen et al., 2016; Schroeders et al., 2022), but the dominant practice—flagging respondents identified by several indices under separately chosen cutoffs—remains informal. A trained classifier that learns the optimal combination of complementary detectors, and that is evaluated honestly across independent data, is the natural next step. This is the motivation for the Random Forest ensemble of Chapter 4.

Third, every detection method ultimately requires a *threshold*, and the field lacks a validated procedure for setting one without ground-truth labels that no analyst possesses in practice (Alfons & Welz, 2024; Ward & Meade, 2023). A calibration strategy that is both deployment-realistic and competitive with label-dependent oracles would make these methods usable by the applied researchers who need them. This is the motivation for the rate-aware calibration evaluated in Chapter 5.

The remaining chapters take up these gaps in turn. Chapter 2 reviews the statistical concepts on which the methods rest—correlation, exploratory factor analysis, the Random Forest classifier, and the evaluation metrics used throughout. Chapter 3 introduces ReReReRe and its underlying coherence principle. Chapter 4 develops the ensemble that combines ReReReRe with the established detectors surveyed above. Chapter 5 applies the ensemble under a deployment-realistic protocol and characterizes when, and how well, it works.

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